Business Question

Is a proposed price for a given room type in a given neighbourhood underpriced, within the expected price range, or overpriced?

Results

After the data cleaning process was completed, the final dataset contained 48,572 Airbnb listings and 18 variables. From that point onward, analysis shifted toward exploration - patterns examined, outliers flagged - all built on this refined collection.

The average listing price was $140.27, while the median price was $105. As low as $10 appeared in records, contrasted sharply by a peak at $999. The standard deviation was $112.90, indicating considerable variation in listing prices across the dataset.

The initial analysis showed that both location and room type have a strong influence on listing prices. Manhattan listings stood out by carrying steeper price tags when compared to areas elsewhere across the city. Entire units - be they full homes or flats - tended to cost more, especially next to private rooms or spaces meant for sharing.

A pattern-finding method based on an Autoencoder helped detect property listings priced outside expected ranges. By studying traits tied to individual ads, it flagged those acting unlike typical market examples. The final Autoencoder model identified 3,287 anomalous listings, representing approximately 6.8% of the dataset. Listings exceeding the anomaly threshold were flagged for further investigation and analysed as potential cases of overpricing, underpricing, or unusual market behaviour.

Key Insights

Patterns emerged during exploration of the Airbnb data. A close look revealed recurring trends worth noting.

One of the most important findings was the impact of neighbourhood location on listing prices. Prices climbed highest in Manhattan when compared across regions, yet stayed furthest down in the Bronx. Though many factors matter, where something is found tilts the scale early.

Figure 1. Price Distribution by Neighbourhood Group

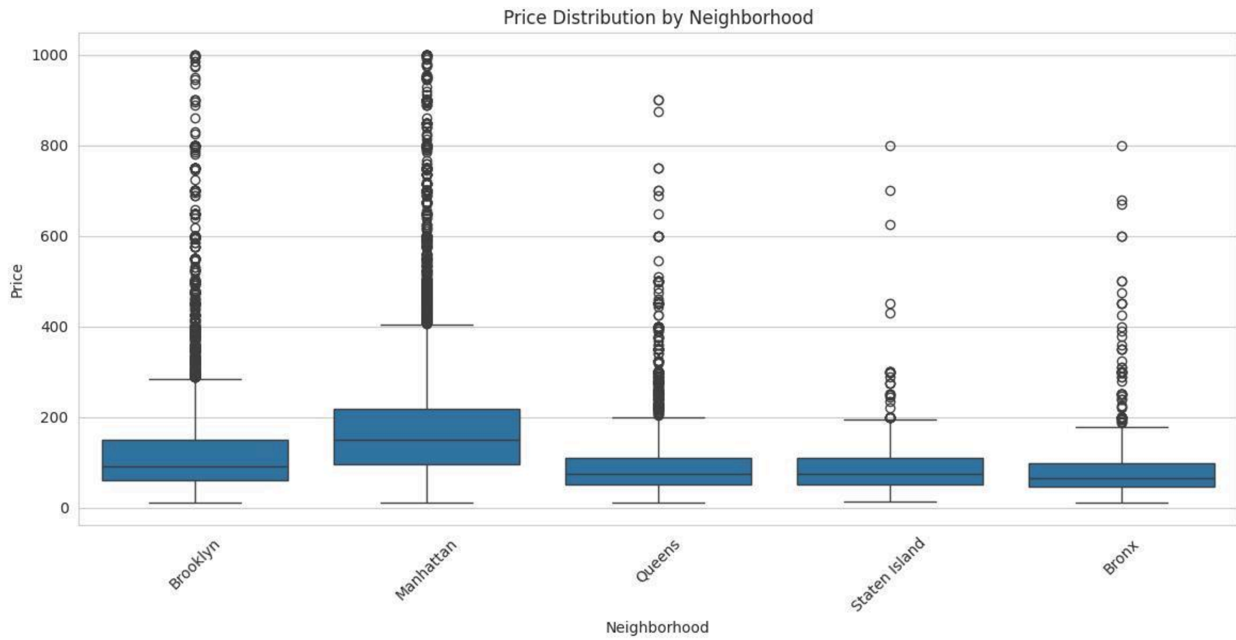

Figure 1 compares listing prices across neighbourhood groups. Manhattan exhibits the highest median prices and the widest price range, while the Bronx shows the lowest overall pricing levels. Several extreme values are also visible across all neighbourhoods.

Room type was also found to be an important factor affecting listing prices.Entire homes and apartments were generally more expensive than private or shared rooms.

The price distribution was not evenly spread across the dataset. A majority of entries clustered toward cheaper and mid-level costs instead. In contrast, only a few stood far above others in cost. Because of those outliers, the mean shifted upward noticeably. This pattern pulled the shape of the graph to the right slightly.

Figure 2. Distribution of Airbnb Prices

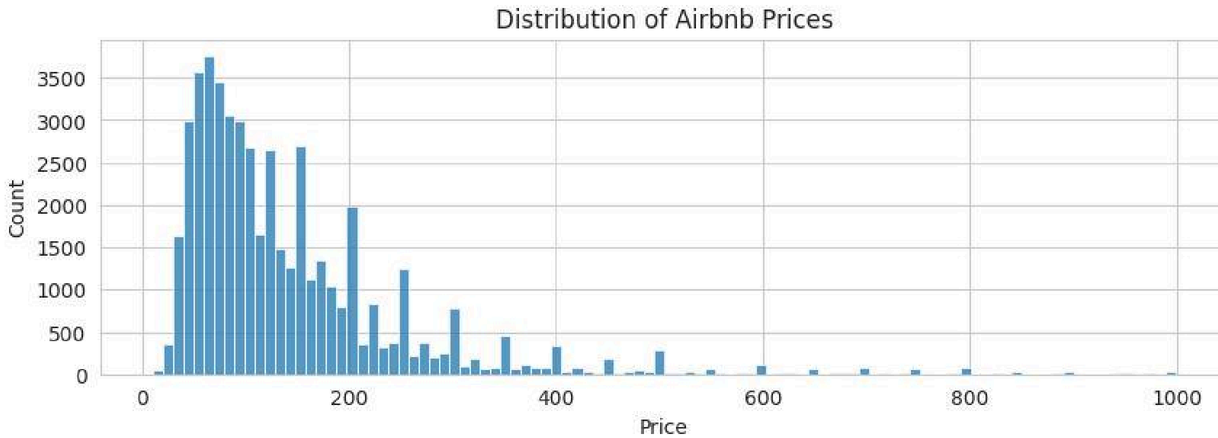

Figure 2 illustrates the distribution of Airbnb listing prices in New York City. The distribution is positively skewed, with most listings concentrated at lower and mid-range prices, while a smaller number of high-priced listings create a long right tail.

Overall, neighbourhood location appeared to be the strongest factor influencing Airbnb prices in New York City. Understanding these patterns helps provide context for identifying unusual pricing behaviour within the dataset.

Patterns Detected

The Autoencoder-based anomaly detection model was trained using key numerical features including price, availability, and review-related variables. Reconstruction error was used as the anomaly score, and listings exceeding the 95th percentile threshold were classified as anomalous.

The model identified 3,287 anomalous listings, representing approximately 6.8% of the analysed Airbnb properties. These listings exhibited characteristics that differed substantially from normal market behaviour and therefore warranted further investigation.

The findings demonstrate that self-supervised learning can effectively identify properties whose characteristics deviate from expected market patterns without requiring pre-labelled examples of anomalous behaviour. This approach provides a practical method for detecting potentially overpriced, underpriced, or otherwise unusual listings within large accommodation datasets.

Top Overpriced Listings

The anomaly detection model identified several listings with exceptionally high prices relative to their feature profiles. These properties produced some of the highest reconstruction errors and were therefore classified as potentially overpriced or highly unusual.

Examples of the most notable overpriced listings included:

• Empire City – King Lux King Room (Price: $999)

• Amazing Views 3BR 2BA Bright and Spacious (Price: $999)

• Beautiful 5100 SQ FT Industrial Brooklyn Loft (Price: $999)

• A Suite with Breathtaking Views of NYC! (Price: $999)

• Luxury Full-Floor 2 Bed Loft with Huge Private Roof (Price: $999)

These listings were primarily located in Manhattan and Brooklyn, where premium accommodation is common. However, the model determined that their overall feature combinations differed significantly from typical listings within the dataset, suggesting that they represent unusually priced properties.

Top Underpriced Listings

The model also detected listings that appeared unusually inexpensive when compared with similar properties. Although low prices do not necessarily indicate errors, they may represent promotional pricing, incomplete listing information, data-entry issues, or genuinely undervalued properties.

Examples of the most notable underpriced listings included:

• Spacious 2-Bedroom Apt in Heart of Greenpoint

• Spacious and Modern 2 Bedroom Apartment

• Studio with Amazing View

• Happy Home 3

• Beautiful Furnished Private Studio with Backyard

These listings displayed characteristics that differed from the pricing patterns typically observed for comparable properties. Their anomaly scores indicate that they deserve further examination to determine whether the pricing reflects market conditions or unusual listing behaviour.

Limitations

One limitation of the current anomaly detection approach is that neighbourhood_group was not included directly as an input feature during Autoencoder training. While this simplified the model and supported the Neural Architecture Search process, it may have reduced the model's ability to identify anomalies relative to local market conditions. Future work could incorporate neighbourhood-specific encoding so that pricing behaviour is evaluated within each geographical area. This may improve the accuracy and interpretability of anomaly detection results.

Conclusion

This project analysed Airbnb listings in New York City to better understand pricing patterns and identify unusual market behaviour. After cleaning and exploring the data, clear patterns began to emerge. Location mattered greatly - listings in Manhattan stood out with steeper rates. Entire homes also tended toward higher costs compared to shared or private rooms. Price variation was tied closely not just to area but also to how much space guests could access.

The findings demonstrate how data analysis and machine learning can be combined to identify meaningful patterns within large datasets. These insights can support better decision-making, improve understanding of market behaviour, and help identify listings that differ from typical pricing trends.

The anomaly detection model successfully identified listings that deviated from normal market patterns, demonstrating the effectiveness of self-supervised learning for anomaly detection in Airbnb pricing data.

Furthermore, all project materials, including the source code, notebooks, and implementation files, have been made available through the project GitHub repository to support transparency, reproducibility, and future development:

https://github.com/Mr-Ratman/ML_team_project

References

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